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c) Reaction with thionyl chloride (Sulphonyl chloride)

Example

$$CH_3CH_2OH + SOCl_2 \xrightarrow[\text{Pyridine}]{} CH_3CH_2Cl + SO_2\uparrow + HCl\uparrow$$

Ethanol → Chloroethane

This reaction is known as **Darzen's halogenation**

2) From alkenes

Alkenes react with halogen acids (HCl, HBr, HI) to give haloalkane. The mode of addition follows Markovnikov's rule.

3) From alkanes

Alkanes react with halogens (Cl_2 or Br_2) in the presence of ultra violet light to form haloalkane. This reaction is a free radical substitution reaction and gives a mixture of mono, di or poly substituted haloalkane.

Example

Chlorination of methane gives different products which have differences in the boiling points. Hence, these can be separated by fractional distillation.

$$CH_4 \xrightarrow[-HCl]{Cl_2 / light} CH_3Cl \xrightarrow[-HCl]{Cl_2} CH_2Cl_2 \xrightarrow[-HCl]{Cl_2} CHCl_3 \xrightarrow[-HCl]{Cl_2} CCl_4$$

Methane → Chloromethane → Dichloromethane → Trichloro methane → Tetrachloro methane

4) Halogen exchange reactions

a) Finkelstein reaction

Chloro or bromoalkane on heating with a concentrated solution of sodium iodide in dry acetone gives iodo alkanes. This reaction is called Finkelstein reaction, (S_N2 reaction).

$$CH_3CH_2Br + NaI \xrightarrow[\Delta]{\text{Acetone}} CH_3CH_2I + NaI$$

Bromoethane → Iodoethane

b) Swarts reaction

Chloro or bromo alkanes on heating with metallic fluorides like AgF, SbF_3 or Hg_2F_2 gives fluoro alkanes. This reactions is called Swarts reaction.

Example

$$CH_3CH_2Br + AgF \xrightarrow{\Delta} CH_3CH_2F + AgBr$$

Bromo ethane → Fluoro ethane

5) From silver salts of fatty acids (Hunsdiccker reaction)

Silver salts of fatty acids when refluxed with bromine in CCl_4 gives bromo alkane

$$CH_3CH_2COOAg + Br_2 \xrightarrow[\text{reflux}]{CCl_4} CH_3CH_2Br + CO_2 + AgBr$$

Silver propionate → Bromo ethane